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Toyota's Air-Fuel Sensing: A look inside an ECU

Horsepower and torque are one of the many things' car enthusiasts strive to squeeze out of an internal combustion engine. It is directly related to phenomena such as Volumetric Efficiency and the combustion quality present in the engine chamber. However, to tweak these parameters, a measurement of the quality of the combustion is necessary to continually tweak and feedback the engine parameters until the desired target values are obtained such as Horsepower and Torque. The Air Fuel Ratio (AFR), or the inverse relation Lambda (λ) is used. This value is used to determine how much air is being used for combustion in relation to the fuel.

$$\boxed{AFR = \frac{m_{air}}{m_{fuel}}}$$

$$\boxed{\lambda = \frac{AFR}{AFR_{stoich}}}$$

a. Equation 1 – Air-Fuel Ratio

b. Equation 2 – Lambda

This value in AFR or Lambda format directly aids in the tuning of combustion engines. There are three states: Lean, Stoichiometry and Rich. Lean is an abundance of air and lack of fuel, whereas Rich is an abundance of fuel and a lack of air; Stoichiometry (or stoich) is when both values are balanced. The AFR value depends on the type of fuel used (methanol, gasoline, diesel, etc.). For gasoline powered engines, the following convention is used:

Table 1:

AFR	Lambda (λ)	Combustion Behavior
12:1	0.8	Rich
14.7:1	1	Stoichiometry
18:1	1.2	Lean

Note: Lambda, regardless of AFR, will follow the convention of rich when less than 1, Stoichiometry at 1 and lean at above 1.

Ideally, the measurement of AFR must have great resolution and accuracy to be measured reliably. In reality, AFR sensors are divided into two categories: Narrowband and Wideband. Narrowband corresponds to an ECU or analog measurement from 0V through 1.2V (note the similarity between Table 1's Lambda values and this voltage range). However, as the name suggests, this leads to measurement being too low in terms of resolution, therefore, no good measurements could be obtained other than to verify if the combustion is at the stoichiometry point or not.

As expected, that does not say a lot about how rich or how lean a combustion actually is. This is where Wideband sensors fix the issue. These sensor types have a wider range present for ECU or analog measurement such as 0V-5V. This higher range now allows someone to determine how an engine responds to combustion, hence, allowing tuning and more importantly, fine tuning for optimizations.

In the 2000s, Toyota had switched their air-fuel measurement system to allow for more resolution. This would allow better performance, emissions and even improved gas mileage. However, their sensors don't follow the previously mentioned example of 0V-5V, rather it does not even utilize voltage to dictate this; the sensor itself behaves as a current source, where if measurement is to be made, it must be powered according to the sensor technology.

The sensor used is Zirconia based. The measurement relies on two electrodes at a differential voltage. Toyota utilizes 300mV and powers the sensor with 3.3V at the anode, and 3.0V at the cathode. This is how the sensor behaves across expected Lambda values:

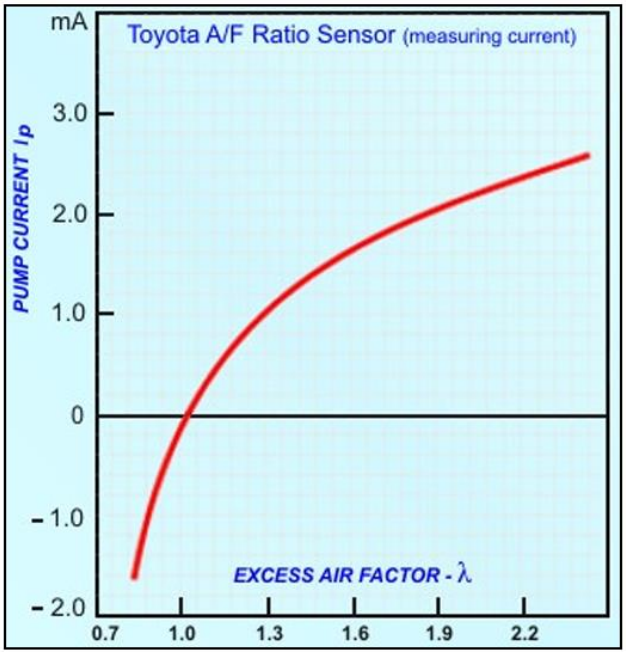


Figure 1 - A/F Ratio Sensor Current vs Lambda Graph

As seen in Figure 1, the current is non-linear, and not magnitude-symmetric. The lean condition can converge up to 3.0mA, and the rich condition can converge to -2.0mA.

Due to the current sensing nature of this application, there is no other alternative for the design but to have a circuit in series with the sensor inputs (signal conditioning):

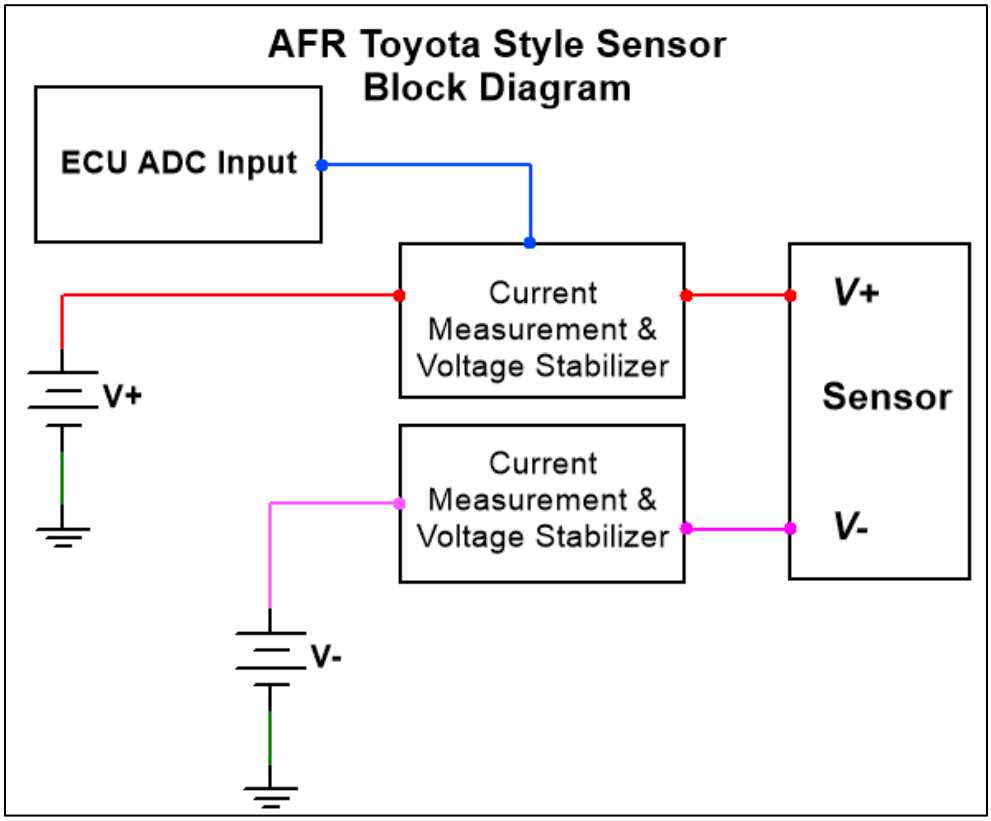


Figure 2 - Toyota Style Sensor Block Diagram

Note: The main current monitoring may be done at one rail only. Toyota utilizes the positive rail, so, the negative lead may be used for DTC/fault detection if built symmetrically.

The ECU Signal Tracing:

To investigate further, I utilized a 2013 Scion TC ECU to attempt to identify and trace the signal path of the AFR sensor. Due to the secret nature of PCB designs, most circuits of interest are hidden beneath proprietary ICs (Integrated Circuits). While hindering our ability to definitively reverse engineer the circuit used by the OEM, we can make educated guesses and infer from what we see, creating a close recreation of what the OEM Circuit might be doing.

As intimidating as the densely packed PCB might seem, continuity does most of the heavy lifting. All that is left is to manually find all present traces of the signal and obtain a netlist or equivalent circuit diagram seen by the inputs.

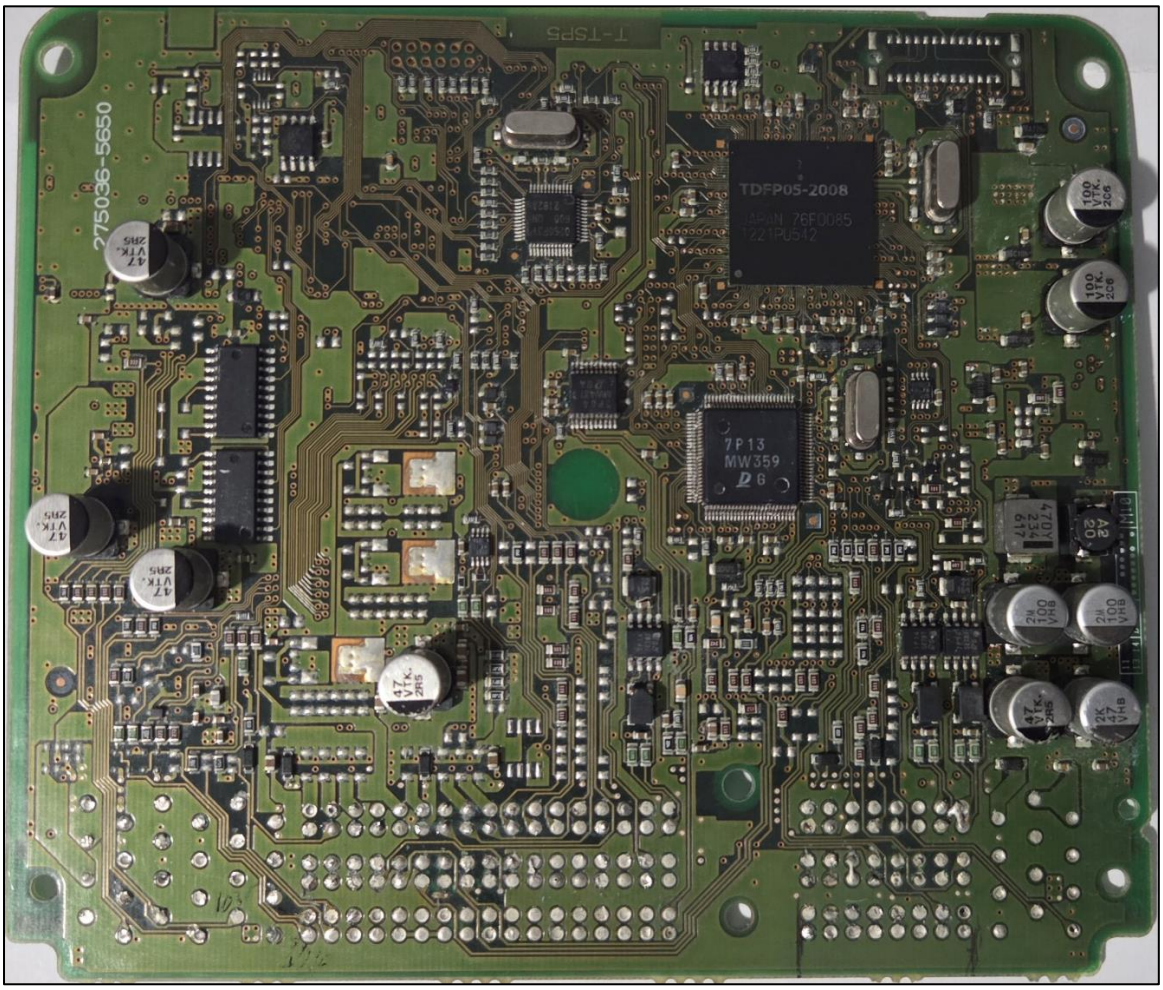


Figure 3 - View of Top layer of ECU

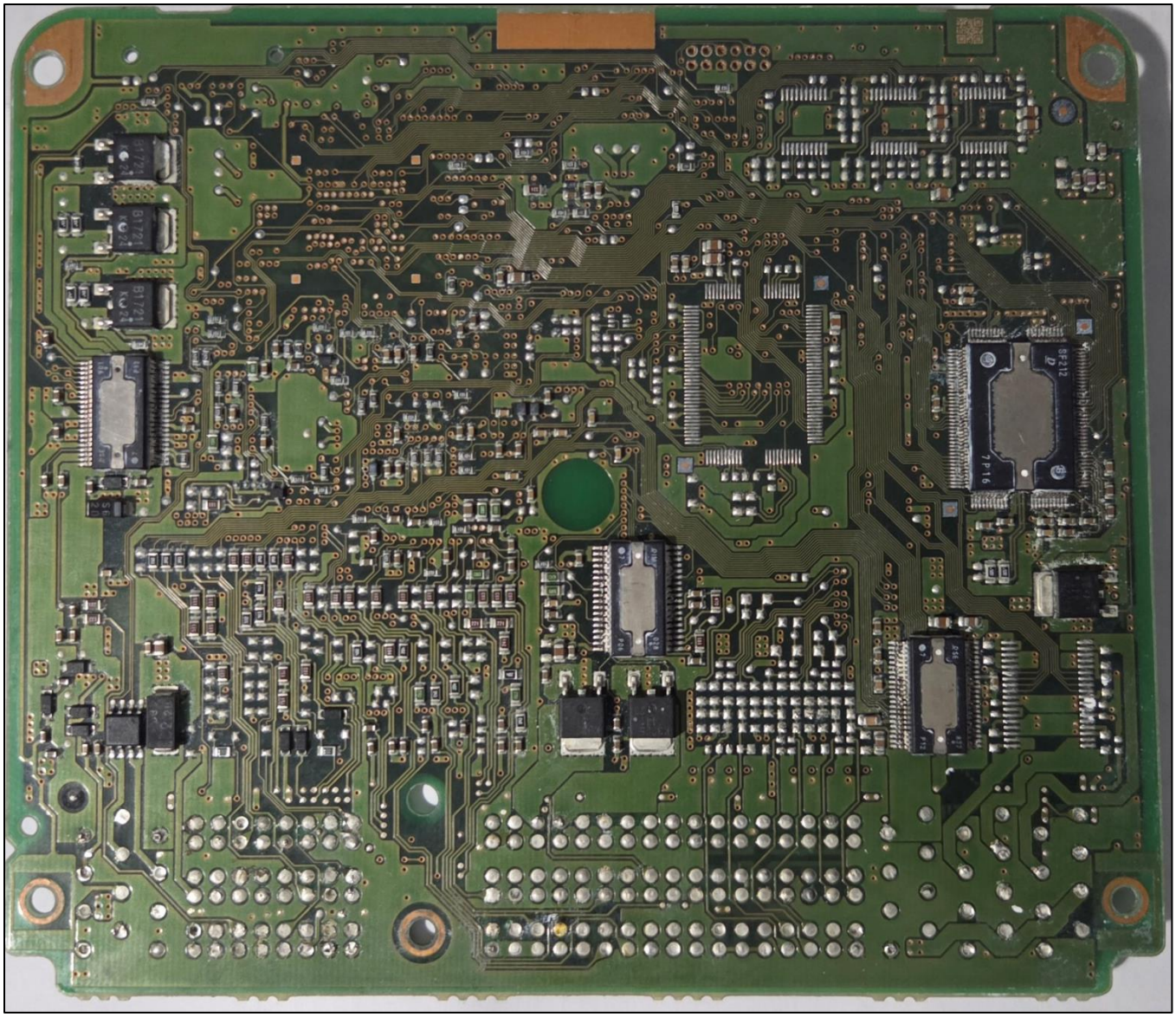


Figure 4 - View of Bottom layer of ECU

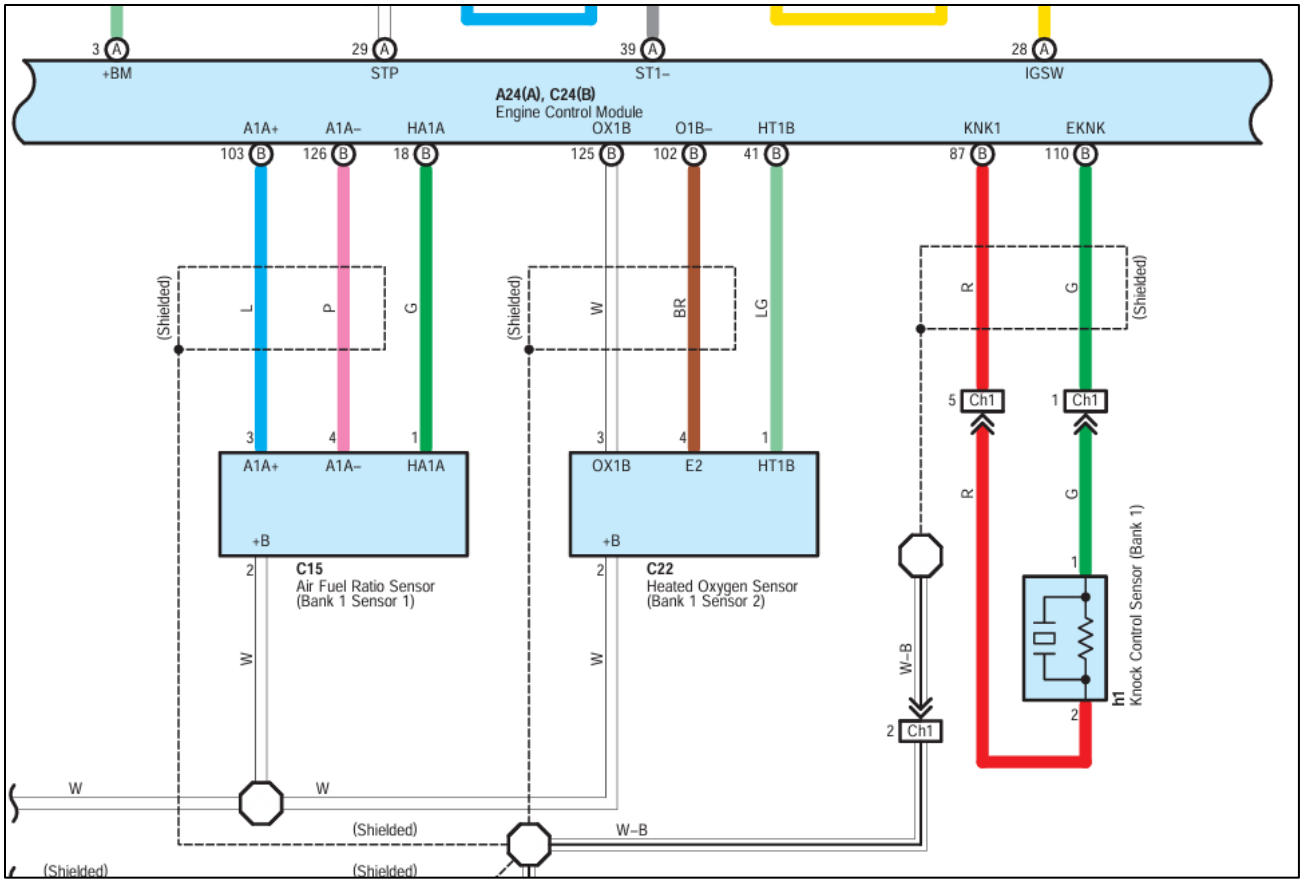


Figure 5 - Electrical Wiring Diagram Excerpt

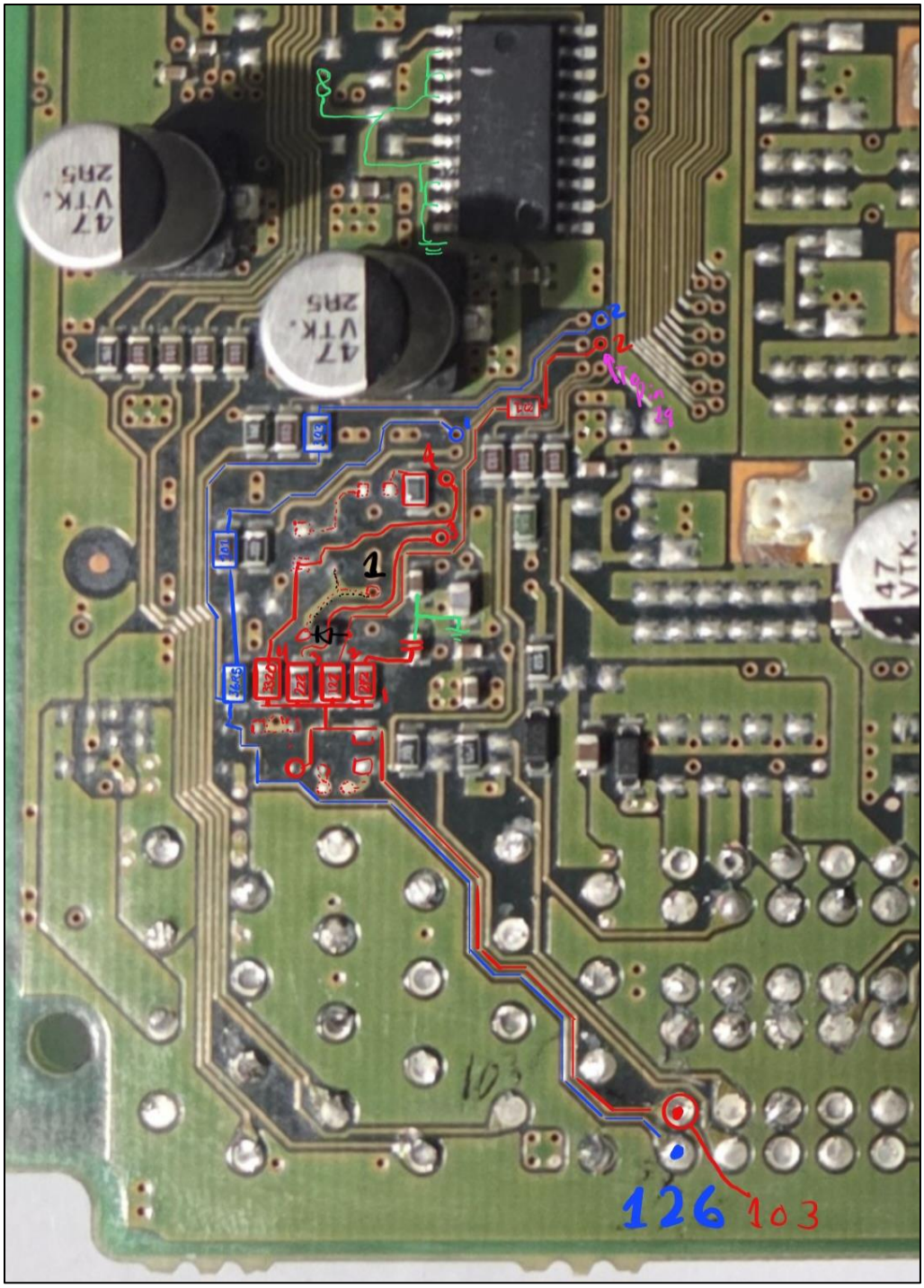


Figure 6 - Signal Tracing

Note: Pin/Pad 126 corresponds to A1A- (Negative sensor rail) and Pin/Pad 103 corresponds to A1A+ (Positive sensor rail) as seen on the Electrical Wiring Diagram.

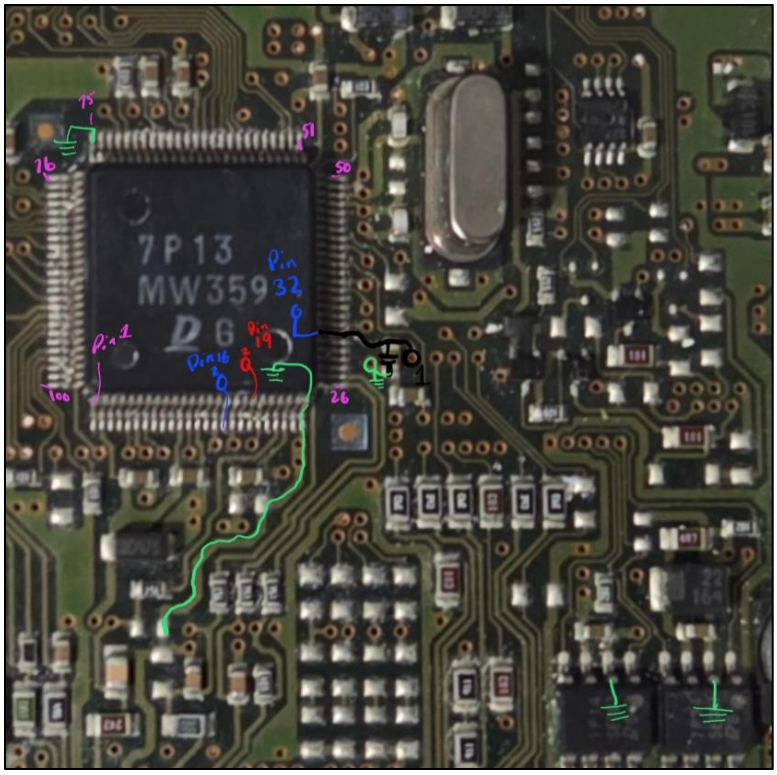


Figure 7 - Signal Tracing

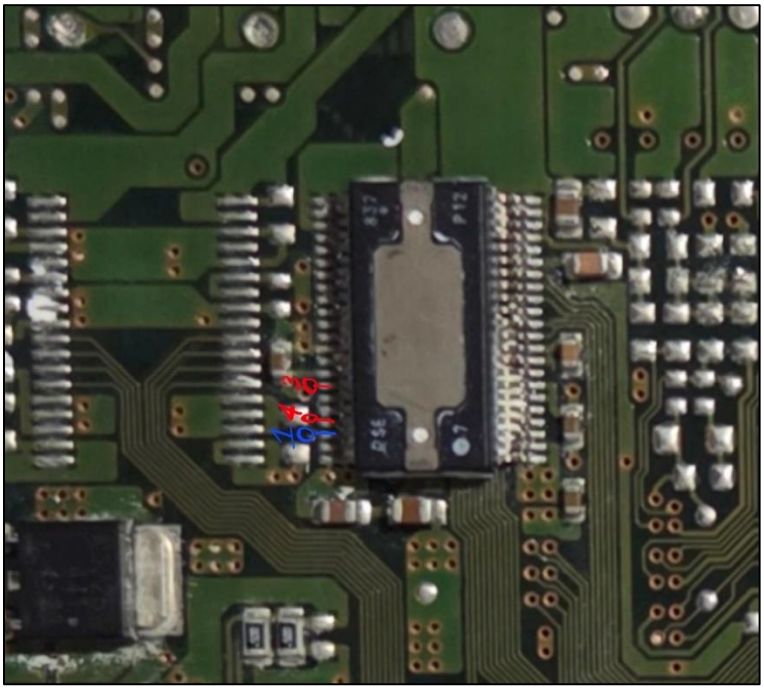


Figure 8 - Signal Tracing

Looking online, there is page with some pertinent info in regard to the Denso SE837 IC's specs and testing. However, this should be taken with a grain of salt:

Parameter	**Symbol**	**Min**	**Typical**	**Max**	**Unit**
Input Voltage	V_{IN}	4.5	-	28	V
Output Voltage	V_{OUT}	1.2	-	20	V
Output Current	I_{OUT}	-	3	5	A
Quiescent Current	I_Q	-	10	20	μA
Efficiency	η	-	90	-	%
Dropout Voltage	$V_{DROPOUT}$	-	1.2	-	V
Operating Temp.	T_{OPR}	-40	-	125	°C
Storage Temp.	T_{STG}	-65	-	150	°C

Table 2: Denso SE837 Product Attributes

From the obtain signal path, a more coherent circuit diagram can be pieced:

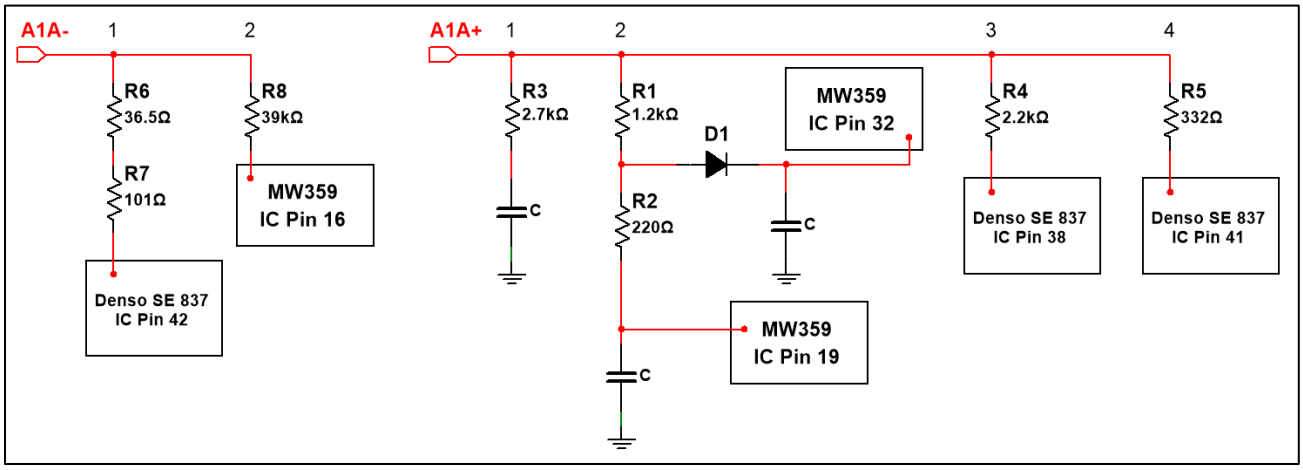


Figure 9 - AFR Sensor PCB Signal Path

As seen, we mostly have two IC's directly interfacing with the Sensor. Nonetheless, whether these two ICs are solely responsible for the measurement or if these ICs connect the internal output elsewhere is hard to say. Since the signal disappears into the ICs in different nodes, then what happens inside and after would mean knowing what is inside the IC. If we consider the previous table and information from the website, then the SE837 IC is some sort of power related IC. This could make sense for driving the circuit. However, the 332Ω resistor seems low enough yet high enough to possibly be shunt resistor.

The second IC, the MW359, seems like a microcontroller chip. It makes sense to read analog values if the pins that are wired are ADC, or Pull-down/Pull-up certain nodes, such as in the second branch of the positive sensor rail signal path.

Conclusion

Through the signal tracing of the AFR sensor in the mentioned ECU, this document intends to provide you, the reader, with a look into the internal circuit path of the air-fuel measurement system. By manually tracing the signal path of the sensor's inputs into the ECU, the work aims to provide some hardware level of understanding despite the limitations imposed by the proprietary ICs.

Ultimately, this paper serves as a contribution to the broader effort of understanding automotive ECU hardware. Whether used for further research, educational circuit analysis, or simply as a reference for those curious about Toyota's AFR sensing implementation, the work presented here is intended to remain useful beyond this investigation.

Works Cited

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